Experiments: Part II

PS200C Quant Methods in Politics, Spring 2026

Chad Hazlett

UCLA

chazlett@ucla.edu

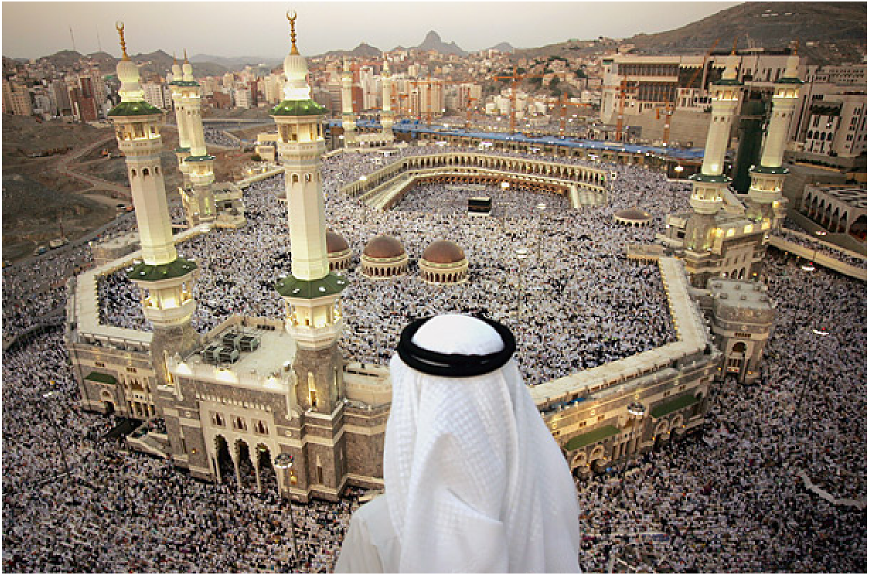
Threats to Internal and External Validity for Experiments

- Internal validity: can we estimate the treatment effect for our particular sample?
  - Fails when there are differences between treated and controls (other than the treatment itself) that affect the outcome
  - In other words, how might the randomization have failed? (See next)
- External validity: can we extrapolate our estimates to other populations?
  - Might Y_{1i}, Y_{0i}, and τ_i have looked different in a different part of the population?

Most Common Threats to Internal Validity

- Randomization fails at assignment
  - e.g. implementing partners assign their favorites to treatment group; imbalance due to small sample size
- Noncompliance with assignment
  - e.g. failure to treat or "crossover". Some members of the control group receive the treatment and some members of the treatment group go untreated
  - e.g. in JTPA: only 62% of women and 66% of men assigned for treatment actually enrolled in JTPA training (on the other side, compliance was almost perfect in the control group).
- Differential attrition
  - attrition *can* be okay, even if different rates for treated and controls
  - but, if probability of attrition depends on treatment status **and** potential outcomes, you break the randomization
  - Two scenarios: (i) people attrite from treatment-taking, but you still get to measure outcome; (ii) people attrite and you can't measure any outcome for them post-treatment.

Example: Clingingsmith, Khwaja, Kremer



Example: Natural Experiment in Pakistan

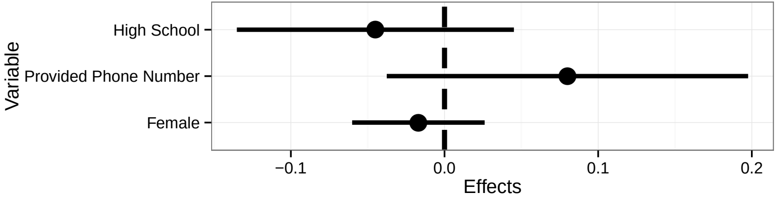
- Pakistan allocated about 135,000 visas to Saudi Arabia for the Hajj via a randomized lottery.
- Wealthier Pakistanis tend to use private Hajj tour operators rather than the lottery.
- “Parties” apply to go, and are randomly selected, with stratification on sect, region, and accommodation.
- Compliance with the experiment is imperfect:
  - 99% who win lottery attend the Hajj.
  - 11% who lose lottery still attend the Hajj (via private tours).

Example: Natural Experiment in Pakistan

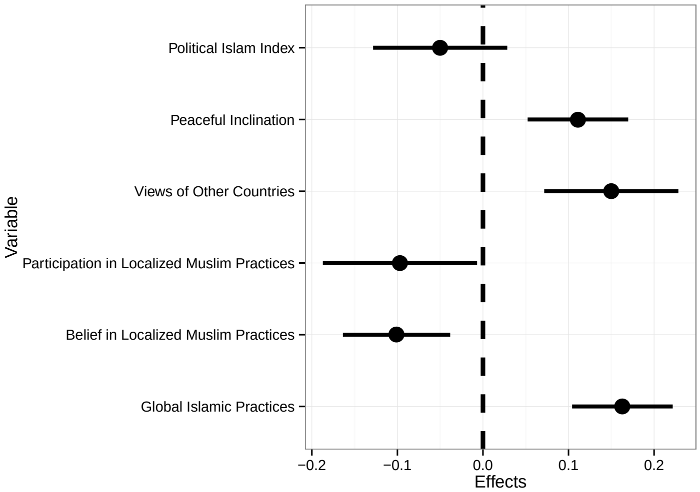
Two kinds of information to bolster the randomization assumption:

1. Qualitative information:
   - The lottery selection algorithm was designed and implemented by an independent and reputable third party, and there were no reports of lottery manipulation.

2. Quantitative information: Balance tests on *observables*



Difference in Means: lottery winner vs. losers



Would you say these are the “effects of going on the Hajj”?

Most Common Threats to External Validity

- Non-representative sample
  - e.g. laboratory experiment using a convenience sample
  - units are randomly assigned, but not from the population of interest
- Non-representative treatment
  - the treatment differs from what you would actually implement
  - e.g. survey experiment about the effect of media priming on voting
  - scale effects
  - actual treatments may be bundled

Internal vs. External Validity

Which one is more important?

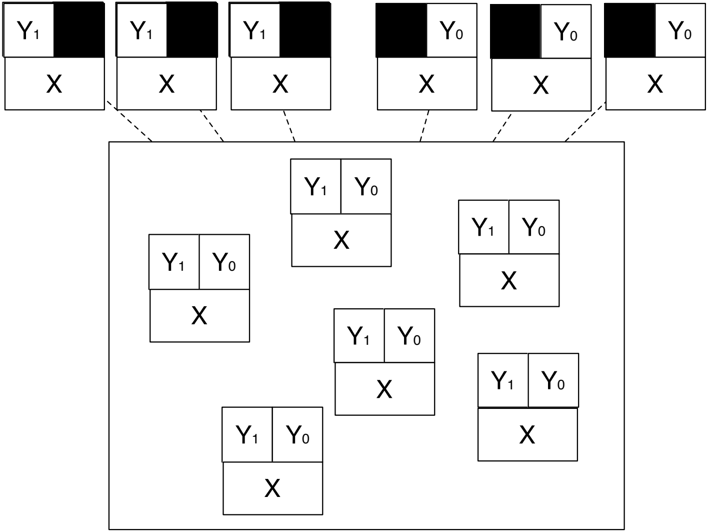
"One common view is that internal validity comes first. If you do not know the effects of the treatment on the units in your study, you are not well-positioned to infer the effects on units you did not study who live in circumstances you did not study." (Rosenbaum 2010, p. 56)

- Randomization (sometimes) ensures internal validity
- External validity may be partially addressed by comparing the results of several internally valid studies conducted in different circumstances and at different times
- While experiments can be especially narrow, same external validity issues often apply in observational work

That said,

- Often studies with lower internal validity give us the theory, inspiration, qualitative knowledge needed for an experiment.
- Our aim is to get away from "valid" and "not valid" to instead discuss the evidence that a given conclusion would hold.

Covariates and Experiments



Covariates

- Randomization is gold standard for causal inference because in expectation it balances observed but also unobserved characteristics between treatment and control group.
- Unlike potential outcomes, you observe baseline covariates for all units.
- We mean "pre-treatment" covariates here, or rather, those not affected by D_i ...Though there are some other interesting cases of covariates you should not include, but that's for another day.
- Randomization implies *covariate balance*,

$$p(X|D=1) = p(X|D=0)$$

- If this is not the case, then one of two possibilities:
  - randomization was compromised.
  - finite sample bad luck
- Good idea to (i) check "balance", and/or (ii) adjust for chance imbalances to at least improve efficiency.

Regression with Covariates

A common model with experimental data:

$$Y_i = \alpha + \tau D_i + X_i\beta + \varepsilon_i$$

Though soon we will consider this (better) model:

$$Y_i = \alpha + \tau D_i + (X_i - \overline{X})\beta + D_i(X_i - \overline{X})\theta + \varepsilon_i$$

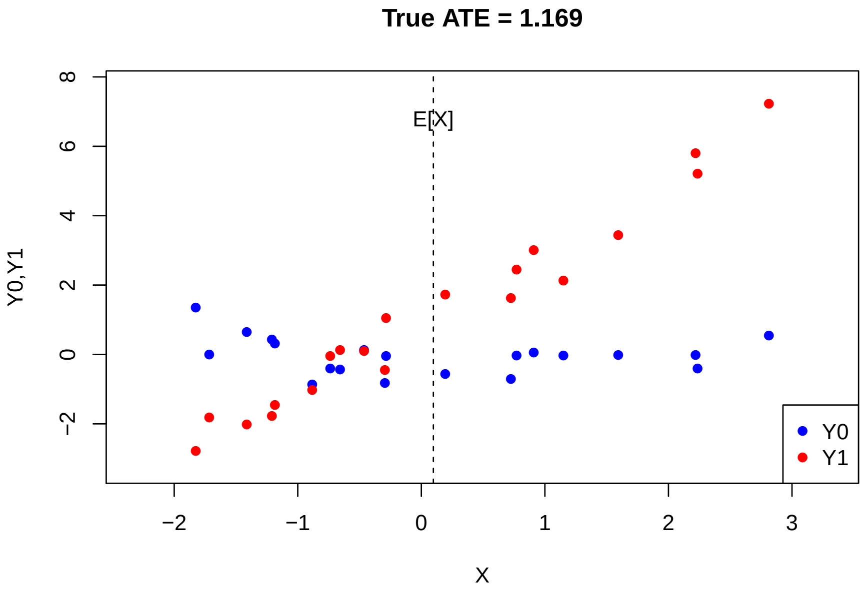
But why include X_i when experiments "control" for covariates by design?

- Correct chance imbalances that may push $\hat{\tau}$ far from τ_{ATE}.
- Increase precision
- (Those are really the same in a given sample)

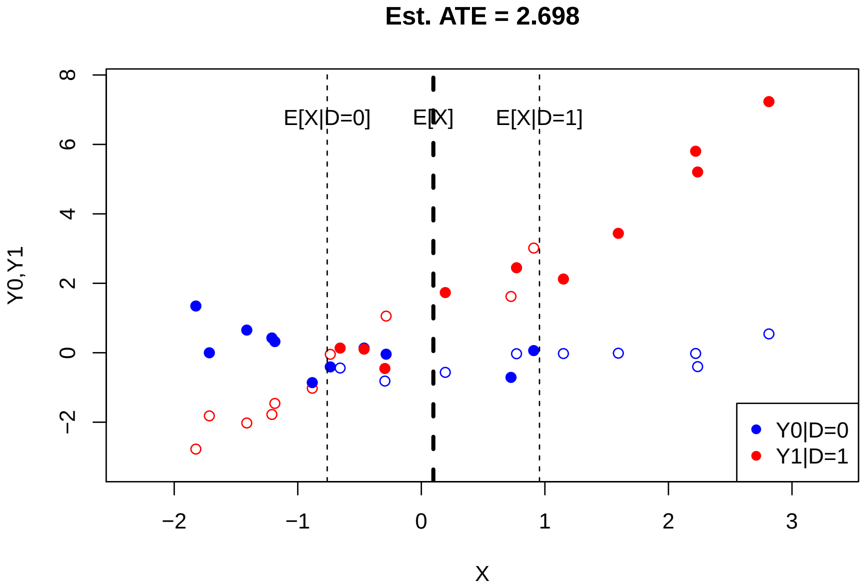
ATE estimates are robust to model specification (with sufficient N).

- But, do not control for **post-treatment** covariates!
- This is an over-simplification: sometimes post-tx variables are okay, sometimes your QOI involves post-tx control, and sometimes pre-tx variables are problematic...we will need DAGs for that.

Regression Adjustment for Experiments

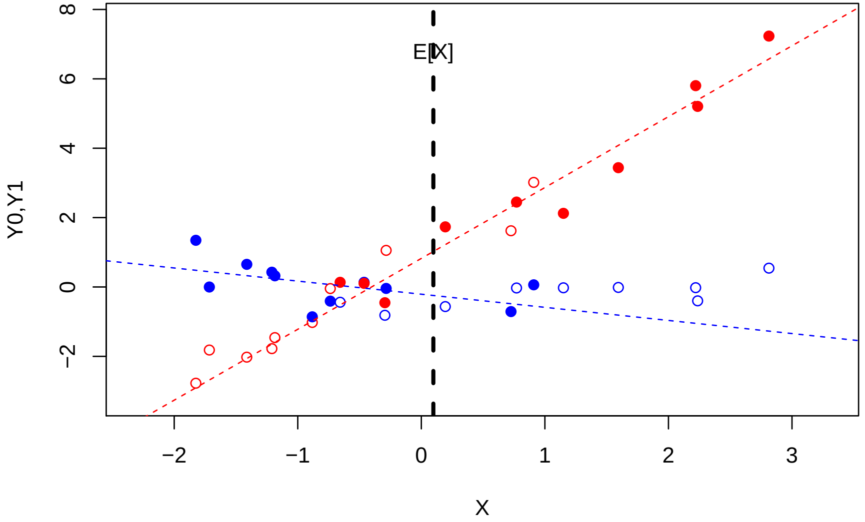


Regression Adjustment For Experiments

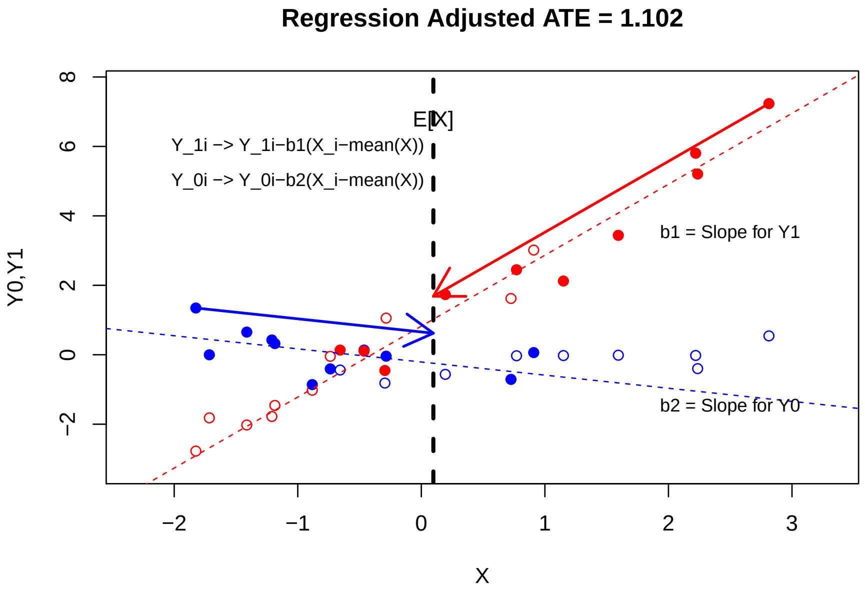


Regression Adjustment For Experiments

Consider regression lines (interacting with D)



Regression Adjustment For Experiments



Covariate Adjustment with Regression

Freedman (2008) noted that regression of the form:

$$Y_i = \alpha + \tau_{reg} D_i + \beta_1 X_i + \varepsilon_i$$

is slightly biased (order $1/N$) in this scenario, does not necessarily improve precision, creates incorrect confidence intervals.

Lin (2013) argued:

- if you interact covariates with D it never hurts and likely improves precision:

$$Y_i = \alpha + \tau_{Lin} D_i + \beta_1 \cdot (X_i - \bar{X}) + \beta_2 \cdot D_i \cdot (X_i - \bar{X}) + \varepsilon_i$$

- this is what we mimicked graphically

Better still, (see this working paper):

- regression with covariates induces a "weird weighting" we will discuss later... turns out that this interactive regression avoids this
- this is also equivalent to some other methods you may hear about (g-computation, t-learner, regression imputation)

Why the weird demeaning business in this model?

It is isomorphic to a model without the demeaning, but makes interpretation for ATE more immediate.

$$Y_i = \alpha + \tau_{Lin} D_i + \beta_1 \cdot (X_i - \bar{X}) + \beta_2 \cdot D_i \cdot (X_i - \bar{X}) + \varepsilon_i$$

Interpreting τ_{Lin}:

- What is $\mathbb{E}[Y_i | D_i = 0, X_i = \overline{X}]$?
- What is $\mathbb{E}[Y_i | D_i = 1, X_i = \overline{X}]$?
- What is $\mathbb{E}[Y_i | D_i = 1, X_i = \overline{X}] - \mathbb{E}[Y_i | D_i = 0, X_i = \overline{X}]$?
- What is $\mathbb{E}[Y_i | D_i = 1, X_i = x^*] - \mathbb{E}[Y_i | D_i = 0, X_i = x^*]$ for $x^* \neq \overline{X}$?

So,

- τ_{Lin} tells you comparison after "moving" all points to $X_i = \overline{X}$
- model could be used to give you $ATE(X)$ but conveniently, $\mathbb{E}_x[ATE(X)] = \tau_{Lin}$.
- allowing this interaction is a good idea, emerging as standard practice

Summary: Covariate Adjustment with Regression

- “Agnostic” view of regression: one does not need to believe in the classical linear model (linearity and constant treatment effects) to advocate OLS covariate adjustment in randomized experiments
- Helpful both to protect against chance imbalance and improve efficiency.
- Since covariates are balanced by design, results typically not model dependent (see appendix for FWL explanation)
- Best reason to be skeptical of adjustment: fishing.
  - always report unadjusted result first
  - best if adjustment strategy pre-specified

Blocking

- Prior to randomization, collect background data on each unit.
- Why leave it to chance to balance on these, especially in small sample?
- Blocking: better-than random balance by design:
  - pre-stratify sample on important factor(s)
  - randomize within each stratum
- Randomization is still there to balance unobserved attributes in expectation
- Why is this helpful?
  - Four subjects with Y_{0i} given by $\{2, 2, 8, 8\}$. (Say Y_{1i} are similar).
  - Simple random assignment into two treated/control will place $\{2, 2\}$ and $\{8, 8\}$ together in the same group one-third of the time

Blocking: Estimation

Imagine you run an experiment where you block on sex as designated at birth.

Obtain block specific ATEs, $\tau_f = \mathbb{E}[\tau|f]$, $\tau_m = \mathbb{E}[\tau|m]$.

Applying the LIE,

$$\mathbb{E}[\tau] = p[f]\mathbb{E}[\tau|f] + p[m]\mathbb{E}[\tau|m]$$

The sample analog is unbiased for this:

$$\hat{\tau}_B = \frac{N_f}{N_f + N_m} \cdot \hat{\tau}_f + \frac{N_m}{N_f + N_m} \cdot \hat{\tau}_m$$

or more generally, if there are J strata or blocks, then

$$\hat{\tau}_B = \sum_{j=1}^{J} \frac{N_j}{N} \hat{\tau}_j$$

Blocking with Regression

Can we treat these blocks as regressors in OLS? Yes, with caution

If Pr(D) is equal in all blocks, can use OLS with block intercepts (fixed effects):

$$Y_i = \beta_0 + \tau D_i + \sum_{j=2}^{J} \beta_j \cdot B_{j[i]} + \varepsilon_i$$

If (empirical) $Pr(D = 1)$ changes by block, OLS differently weights the blocks.

We can correct for this by going back to the Lin-style (interacted) regression!

- Again see this working paper
- Another proposal is to weight each observation so that block-treatment probabilities become the same. Let

$$p_{ij} \equiv Pr(D_i = 1 | \text{ block for unit i})$$

$$w_{ij} = \left(\frac{1}{p_{ij}}\right) D_i + \left(\frac{1}{1 - p_{ij}}\right)(1 - D_i)$$

...which turns out to be exactly the same thing as the interactive regression

Fair Trade Labeling Experiment (Hainmueller et al, 2012)

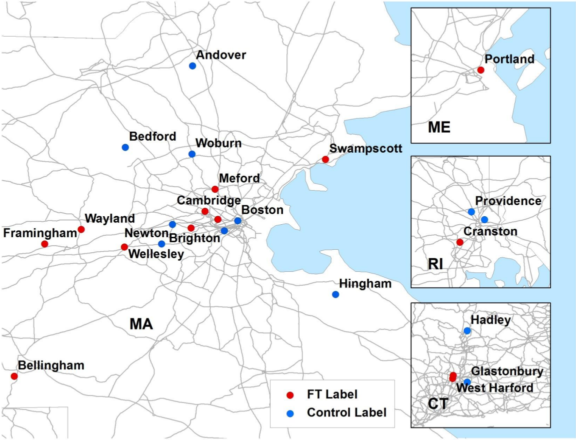
Label Experiment

Treatment

Control

Example: Fair Trade Labeling Experiment

Matched Pairs: Phase 1



Example: Fair Trade Labeling Experiment

```
> d <- read.dta("FTdata.dta")
> head(d)
  store pair FTweek lnsalesd
1     1    1      1     3.20
2     4    1      0     2.77
3     6    2      1     4.18
4     9    2      0     4.04
5    21    3      1     4.30
6    24    3      0     3.93
```

Example: Fair Trade Labeling Experiment

```
> cr.out <- lm(lnsalesd~FTweek,data=d)
> coeftest(cr.out,vcov = vcovHC(cr.out, type = "HC1"))

t test of coefficients:

            Estimate Std. Error t value Pr(>|t|)
(Intercept)  4.35000    0.16079 27.0537   <2e-16 ***
FTweek       0.12385    0.21424  0.5781   0.5686
--
```

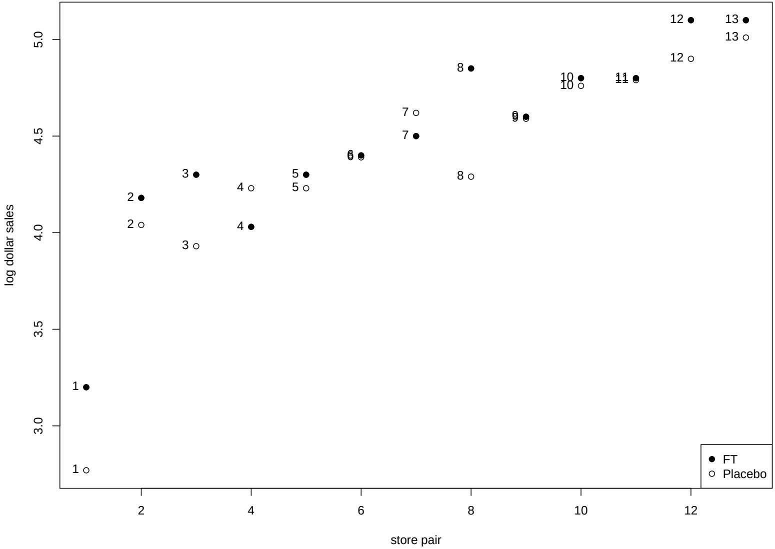
Example: Fair Trade Labeling Experiment

```
> br.out <- lm(lnsalesd~FTweek+as.factor(pair),data=d)
> coeftest(br.out,vcov = vcovHC(br.out, type = "HC1"))

t test of coefficients:

                   Estimate Std. Error t value  Pr(>|t|)
(Intercept)        2.923077   0.162144 18.0277 4.671e-10 ***
FTweek             0.123846   0.060176  2.0581 0.0619840
as.factor(pair)2   1.125000   0.159549  7.0511 1.335e-05 ***
as.factor(pair)3   1.130000   0.204440  5.5273 0.0001304 ***
as.factor(pair)4   1.145000   0.231925  4.9369 0.0003439 ***
as.factor(pair)5   1.280000   0.161773  7.9123 4.208e-06 ***
as.factor(pair)6   1.410000   0.169987  8.2948 2.591e-06 ***
as.factor(pair)7   1.575000   0.203689  7.7324 5.317e-06 ***
as.factor(pair)8   1.585000   0.277319  5.7154 9.675e-05 ***
as.factor(pair)9   1.610000   0.169987  9.4713 6.420e-07 ***
as.factor(pair)10 1.795000   0.165195 10.8660 1.450e-07 ***
as.factor(pair)11 1.810000   0.169987 10.6479 1.810e-07 ***
as.factor(pair)12 2.015000   0.164183 12.2729 3.763e-08 ***
as.factor(pair)13 2.070000   0.160298 12.9134 2.127e-08 ***
--
```

Example: Fair Trade Labeling Experiment



Conclusions: Blocking

How does blocking help?

- Increases efficiency if the blocking variables predict outcomes (see appendix for the variance argument)
- By design, reduced finite-sample bias due to “bad” randomization
- Powerful – can save your experiment!

What to block on?

- Two constraints: data you can get, number of units to work with.
- Irrelevant variables: burn up DOFs
- Most important: predictors of potential outcomes!
- Pre-Tx outcomes is great; add other main predictors if available/helpful
- Another consideration: variables desired for subgroup analysis

How to block?

- Stratification
- Pair-wise matching; two pair-matching

Analysis with Blocking

"**As ye randomize, so shall ye analyze**" (Senn 2004)

- Account for the method of randomization when doing analysis
- If using OLS,
  - include block dummies
  - typically use clustered SEs
  - if probability of treatment assignment varies across blocks, reweight
- If doing randomization inference, block design influences construction of Ω: re-randomization is within blocks.
- Also, type of SE can influence type of bootstrap, e.g. use the block bootstrap (resample at level of blocks) for clustered SEs

Hard work in design and implementation

Statistics are often easy; challenge is in implementation and design

- Finding partners, manage relationships will be common failure point
- Iron law of field experiments: keep it simple. Many ways to fail.
- Getting difficult to publish studies that only examine the efficacy of some intervention: look for substantively interesting treatment and outcome.

Conclusion: Experiments

- Random assignment solves the identification problem for causal inference based on minimal assumptions that the researcher can control
- Put differently, random assignment “balances observed and unobserved confounders”, or “makes treated and control units comparable”.
- Regression *can* be used for analyzing experiments: simple regression with robust SE yields unbiased estimates with conservative confidence intervals.
- Design: blocking to improve balance and efficiency
- Analysis: covariate adjustment (Lin 2013 interactive model) can improve efficiency; relatively safe but pre-specify
- Possible tradeoff between internal validity and external validity

Appendix

Frisch-Waugh-Lovell (FWL) Theorem

Two facts:

1. *any* bivariate regression of some Y on X has $\hat{\beta} = \frac{\widehat{\text{Cov}}(Y,X)}{\widehat{\text{Var}}(X)}$.
2. the coefficient on the kth variable, X_k, from a multivariate regression can be obtained by (i) regress X_k on other X's, call residual $\tilde{X}_k$.(ii) regress Y on $\tilde{X}_k$. (Or $\tilde{Y}$ on $\tilde{X}_k$ for similarly defined $\tilde{Y}$)

All together, this gives the Frisch-Waugh-Lovell (FWL) theorem:

$$\beta_k = \frac{Cov(\tilde{Y}_i, \tilde{X}_{ki})}{Var(\tilde{X}_{ki})}$$

Where,

- $\tilde{X}_{ki}$ is the residual from a regression of X_{ki} on all other covariates
- $\tilde{Y}_i$ may have all other X partialled out as well or you can use the original Y_i.

Why are Experimental Findings Robust to Alternative Specifications?

If our "X_k" of interest is the D:

$$\hat{\beta}_D = \frac{\widehat{\mathrm{Cov}}(\tilde{Y}_i, \tilde{D}_i)}{\widehat{\mathrm{Var}}(\tilde{D}_i)}$$

where $\tilde{D}_i$ is the residual part of D_i after regressing on X_i

- For experimental data, on average, what will $\tilde{D}_i$ look like?
- What does this suggest about the effect of adding covariates to the simple regression of Y on D?

When does blocking in the analysis help?

Imagine models for complete and block-randomized designs:

$$Y_i = \alpha + \tau_{CR} D_i + \varepsilon_i \tag{1}$$

$$Y_i = \alpha + \tau_{BR} D_i + \sum_{j=2}^{J} \beta_j B_{j[i]} + \varepsilon_i^* \tag{2}$$

where $B_j[i]$ is a dummy for the j-th block. Then given iid sampling:

$$Var[\widehat{\tau}_{CR}] = \frac{\sigma_\varepsilon^2}{\sum_{i=1}^n (D_i - \bar{D})^2} \quad \text{with } \widehat{\sigma}_\varepsilon^2 = \frac{\sum_{i=1}^n \widehat{\varepsilon}_i^2}{n-2} = \frac{SSR_{\widehat{\varepsilon}}}{n-2}$$

$$Var[\widehat{\tau}_{BR}] = \frac{\sigma_{\varepsilon^*}^2}{\sum_{i=1}^n (D_i - \bar{D})^2 (1 - R_D^2)} \quad \text{with } \widehat{\sigma}_{\varepsilon^*}{}^2 = \frac{\sum_{i=1}^n \widehat{\varepsilon_i^*}^2}{n-k-1} = \frac{SSR_{\widehat{\varepsilon^*}}}{n-k-1}$$

where R_D^2 is R^2 from regression of D on all block indicators and a constant.

When does blocking in the analysis help?

$$Var[\widehat{\tau}_{CR}] = \frac{\sigma^2_{\varepsilon}}{\sum_{i=1}^{n}(D_i - \bar{D})^2} \quad \text{with } \widehat{\sigma}^2_{\varepsilon} = \frac{\sum_{i=1}^{n} \widehat{\varepsilon}_i^2}{n-2} = \frac{SSR_{\widehat{\varepsilon}}}{n-2}$$

$$Var[\widehat{\tau}_{BR}] = \frac{\sigma^2_{\varepsilon*}}{\sum_{i=1}^{n}(D_i - \bar{D})^2(1-R_j^2)} \quad \text{with } \widehat{\sigma}^2_{\varepsilon*} = \frac{\sum_{i=1}^{n} \widehat{\varepsilon}_i^{*2}}{n-k-1} = \frac{SSR_{\widehat{\varepsilon*}}}{n-k-1}$$

where R_D^2 is R^2 from regression of D on all B variables and a constant.

- If R_D^2 is high, variance inflated. But what can we say about R_D^2 when we're talking about blocking?
- If blocks explain outcomes strongly, how will $\widehat{\sigma}^2_{\varepsilon*}$ compare to $\widehat{\sigma}^2_{\varepsilon}$?
- Adding DOF adjustment, we end up asking whether $\frac{SSR_{\widehat{\varepsilon*}}}{n-k-1} < \frac{SSR_{\widehat{\varepsilon}}}{n-2}$

How much is explained by blocks? (Fair Trade Example)

```
> summary(lm(lnsalesd~as.factor(pair),data=d))
Coefficients:
                  Estimate Std. Error t value Pr(>|t|)
(Intercept)         2.9850     0.1212  24.621 2.72e-12 ***
as.factor(pair)2    1.1250     0.1715   6.562 1.82e-05 ***
as.factor(pair)3    1.1300     0.1715   6.591 1.74e-05 ***
as.factor(pair)4    1.1450     0.1715   6.678 1.52e-05 ***
as.factor(pair)5    1.2800     0.1715   7.466 4.73e-06 ***
as.factor(pair)6    1.4100     0.1715   8.224 1.65e-06 ***
as.factor(pair)7    1.5750     0.1715   9.186 4.77e-07 ***
as.factor(pair)8    1.5850     0.1715   9.245 4.44e-07 ***
as.factor(pair)9    1.6100     0.1715   9.390 3.71e-07 ***
as.factor(pair)10   1.7950     0.1715  10.469 1.05e-07 ***
as.factor(pair)11   1.8100     0.1715  10.557 9.56e-08 ***
as.factor(pair)12   2.0150     0.1715  11.752 2.68e-08 ***
as.factor(pair)13   2.0700     0.1715  12.073 1.94e-08 ***
--
Residual standard error: 0.1715 on 13 degrees of freedom
Multiple R-squared:  0.9474,       Adjusted R-squared:  0.8988
F-statistic:  19.5 on 12 and 13 DF,  p-value: 2.356e-06
```